CHAPTER 4

# Differentiation: from the rate of change to the tangent

**Chapter path**

**Objectives.** Interpret a rate; approximate a derivative; find a tangent; understand failures of differentiability.
**Prerequisites.** Images, slope of a line, algebraic manipulation, square root and absolute value.
**Yearly reference point.** 3·12 November: 8 a.m.

## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Slope of the line through (1; 2) and (3; 8)?

2. Expand $(2 + \quad h)^2 - 4$.

3. Simplify $\dfrac{4h + h^2}{h}$ if $h \neq 0$.

4. Calculate $\sqrt{0}$ and $\sqrt{0{,}01}$.

5. Calculate $|-3|$ and $|2|$.

6. A line with slope 2 passes through (1; 5). Equation?

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a question similar, then explain the rule used.

**Oral work and logic.** Explain to a classmate why we simplify the rate before letting h tend to 0. Distinguish average and instantaneous speed.

## Investigate before formalising

For $f(x) = x^2$, vary h, positive and negative, around 0, keeping point A at x = 1.
Draw three secants; conjecture a limiting line, then determine it without reading the graph.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min.
Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

## 4.1 Rates of change and secants

**Definition and results**

For $a \neq b$ in the domain of f, the rate of change between a and b is $\dfrac{f(b) - f(a)}{b - a}$. This is the slope of the secant through the corresponding points. In context, the unit of the rate is the unit of f divided by that of the variable.

**Status and justification.** Definition; formula for the slope of a line already studied. With $b = a + h$, $h \neq 0$, the rate becomes $\dfrac{[f(a + h) - f(a)]}{h}$.

**Worked example.** For $f(x) = x^2$, between 1 and 3, the rate is $\dfrac{9 - 1}{3 - 1}$ = 4. Between 1 and 1 + $h$, it equals $2 + h$. The slopes for $h = 0{,}1$ and $h = -0{,}1$ are 2.1 and 1.9.

**Watch out**

The rate is not $\dfrac{f(b)}{f(a)}$ ; h is a gap, positive or negative according to the domain.

**Immediate application.** Calculate the rate of $f(x) = x^2 + 3x$ between 2 and 2 + $h$.

## 4.2 Derivative at a point

**Definition and results**

f is differentiable at a when the rate $\dfrac{[f(a + h) - f(a)]}{h}$ tends to a real number as h tends to 0, in the domain. This real number is $f\prime(a)$. An intuitive approach to limits is used here. In kinematics, if f gives a position in metres and time in seconds, the derivative represents an instantaneous speed in m s $^{-1}$.

**Status and justification.** For $f(x) = x^2$, the rate at a equals 2 $a + h$, which approaches 2 $a$ ; so $f\prime(a) = 2a$. This calculation will reappear as a derivative function in Chapter 6.

**Worked example.** For $s(t) = t^2 + 3t$, the rate around 2 is 7 + $h$, therefore $s\prime(2) = 7\ \mathrm{m\,s^{-1}}$. The average speed on [2; 3] is 8 m s $^{-1}$ ; it is not the velocity at time 2.

**Watch out**

We do not replace h by 0 in an unsimplified quotient: we would get a division by 0.

**Immediate application.** For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate.

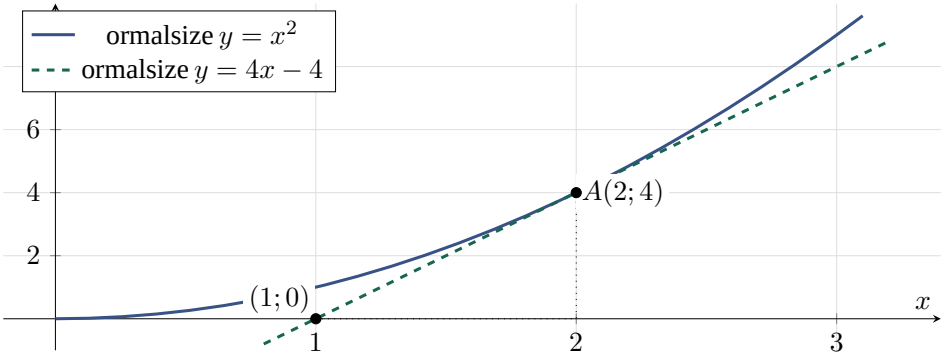
## 4.3 Tangent and linear approximation

**Definition and results**

If f is differentiable at a, the tangent at the point ( $a; f(a)$) has equation $y = f(a) + f\prime(a)(x - a)$. For small h, we use $f(a + h) \approx f(a) + f\prime(a)h$. This local approximation does not give on its own a bound on the error.

**Status and justification.** The tangent has slope $f\prime(a)$ and passes through $(a; f(a))$. In $y = f\prime(a)x + p$, substituting this point gives $p = f(a) - af\prime(a)$, hence the equation.

**Worked example.** For $f(x) = x^2$ at 2, tangent $y = 4 + 4(x - 2) = 4x - 4$. We approximate 2 ,01$^2$ by 4 + 4(0 ,01) = 4,04; the exact value is 4 ,0401, gap 0 ,0001 $= h^2$.




The tangent at A also passes through (1; 0). Its slope is $\frac{4-0}{2-1}$ = 4. The dashed drawing lets us distinguish from the curve, even in black and white.

**Watch out**

A tangent can cross the curve again elsewhere. "A single common point" is not a definition.

**Immediate application.** We are given $f(3)$ = 5 and $f\prime(3) = -2$. Write the tangent and approach $f(3{,}04)$.

## 4.4 When differentiability fails

**Definition and results**

The square root function is defined on [0; + ∞[ but is not differentiable at 0. The function absolute value is defined on $\mathbb{R}$ but is not differentiable at 0. Being defined or having a curve continuous therefore does not guarantee differentiability.

**Status and justification.** For $h > 0$, $\dfrac{\sqrt{h} - 0}{h} = \dfrac{1}{\sqrt{h}}$, which does not tend to a finite real number as h approaches 0. For $|x|$, the rate $\dfrac{|h|}{h}$ is 1 on the right and $-1$ on the left: no common value.

**Worked example.** For $|x|$, the secants through the origin all have slope 1 on the right and $-1$ à left. The corner point does not have a unique tangent with finite slope. For the root, the limiting direction is vertical; there is no real derivative.

**Watch out**

A very steep gradient is not proof of non-differentiability; we must explain the behaviour of the rate.

**Immediate application.** Justify that $g(x) = |x - 2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \overline{x}$ at 0 by its rate of change.

## Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher chooses one. Exercise 8 is revisited later.

### Exercise 4.1 · Two secants

The function $f$ is defined on $\mathbb{R}$ by $f(x) = x^2 - 1$. Let $A$ the point on its curve with x-coordinate 1.

1. Calculate the slope of the secant joining $A$ at the point on the curve with x-coordinate 2.

2. Repeat the calculation with the point of the curve with x-coordinate 1 ,1.

3. After studying the derivative at a point, determine $f\prime(1)$ and compare this value to the two calculated rates. Explain what changes when the second point approaches $A$.

### Exercise 4.2 · One tangent

Consider the function $f(x) = 2x^2 + 1$, defined on $\mathbb{R}$. The aim is to determine the equation of its tangent at the point with x-coordinate $-1$, starting from the rate of change.

1. For $h \neq 0$, write then simplify $\dfrac{f(-1 + h) - f(-1)}{h}$.

2. Deduce $f\prime(-1)$. Also calculate $f(-1)$.

3. Write the equation y = mx + p of the tangent and check that it passes through the point of contact.

### Exercise 4.3 · Marginal cost

A company uses the model $C(q) = q^2 + 20q + 100$, where $q \geq 0$ denotes the quantity produced and $C(q)$ its cost in monetary units. The model extends the quantity to a real variable in order to study small variations in production.

1. Calculate $C(10)$ and $C(11)$, then the rate of change of $C$ between 10 and 11. Interpret this rate in the context.
2. From the rate between 10 and 10 + $h$, determine $C'(10)$.
3. Compare the two results. Explain why the cost of one extra unit is not exactly equal to the derivative in this model.

### Exercise 4.4 · Read a tangent

This exercise has two independent situations. A tangent to a curve is a line: a other point of this line does not necessarily lie on the curve.

1. On the figure in the section on the tangent, read the coordinates of two points on the line green. Deduce its slope and check the value stated in the legend.
2. In another situation, the tangent to the graph of a function $f$ at the point $A(2; 3)$ passes through $B(4; 2)$. Determine $f(2)$ and $f'(2)$. Can we deduce $f(4)$ ?
3. Use this tangent to give an approximate value of $f(2{,}1)$. Show the sign « approximately equal" in the conclusion.

### Exercise 4.5 · Checking an approximation

We want to approximate $1{,}98^2$ without first carrying out the multiplication. We use the tangent to the curve of the square function at the point with x-coordinate 1.

1. Write the equation of this tangent and deduce the desired approximation.
2. Then calculate the exact value of $1{,}98^2$. The error is defined as « exact value minus approximate value approximation": calculate this error and state its sign.
3. Explain the result from the identity $(1 + h)^2 = 1 + 2h + h^2$.

### Exercise 4.6 · Tangent and intersection

The function $f$ is defined on $\mathbb{R}$ by $f(x) = x^3$. We accept that $f'(1) = 3$. A student claims that a tangent and a curve always have exactly one point in common.

1. Determine the equation of the tangent to the graph of $f$ at the point with x-coordinate 1.
2. Check by expanding that $x^3 - 3x + 2 = (x - 1)^2(x + 2)$.
3. Determine all points of intersection of the curve and this tangent. The pupil's claim is it correct?

### Exercise 4.7 · Numerical secants

To study the square function near 1, we define, for $h \neq 0$,

$$T(h) = \frac{(1+h)^2 - 1}{h}.$$

The six values 0 ,1, 0,01, 0,001 and their opposites.

1. Write a Python function that calculates $T(h)$, then produce a table of six results.
2. Conjecture the value of the derivative of the square function at 1.
3. Simplify algebraically $T(h)$ to justify the conjecture. Explain why testing values numerical checks is not enough, on its own, to prove the result.

### Exercise 4.8 · Delayed review

The reciprocal function is defined by $f(x) = \frac{1}{x}$ for $x \neq 0$. It is studied near 1.

1. State the forbidden values of $h$ in the rate $\frac{f(1+h) - f(1)}{h}$, then simplify it.
2. Deduce $f'(1)$ and the equation of the tangent at 1.
3. Use this tangent to approximate $\frac{1}{1{,}02}$. Keep the distinction between exact value and approximate value approximate.

**Independent work.** Do only the task assigned at the end of the session, in the scheme of work Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not use up the whole reserve.

## To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the personal work planned that day (10 to 20 minutes in total).

**CQFD, Exercise 17 p. 127, all questions.** Read two points on each tangent; write the difference quotient rather than estimating its slope.

**CQFD, Exercise 19 p. 127, Questions 1 and 2.** Distinguish the y-coordinate of the point of contact from the slope of the tangent line.

## MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. Rate of $x^2$ between 1 and 2:

A. 2
B. 3
C. 1
D. 4

2. If $f(1) = 4$, $f'(1) = 2$, tangent:

A. $y = 2x + 2$
B. $y = 4x + 2$
C. $y = 2x + 4$
D. $y = x + 3$

3. The derivative at this point is

A. a function
B. an area
C. an x-coordinate
D. a real number if the derivative exists

4. For $x^2$ at 3, derivative:

A. 9
B. 3
C. 6
D. 12

5. The approximation of $f(a + h)$ is

A. $f(a)f'(a)$
B. $f(a) + hf'(a)$
C. $f(a) + f'(h)$
D. $hf(a)$

## MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $\sqrt{x}$ at 0 is

A. differentiable with derivative 0
B. differentiable with derivative 1
C. not differentiable
D. undefined

2. $\dfrac{|h|}{h}$ for $h < 0$ is

A. $-1$
B. 1

C. $0$

D. $h$

3. A tangent with slope 0 is

A. vertical

B. oblique

C. impossible

D. horizontal

4. Position in m and time in s: unit of the derivative

A. $\mathrm{m}^2$

B. $\mathrm{m\,s}^{-1}$

C. $\mathrm{s\,m}^{-1}$

D. $\mathrm{m}$

5. For f differentiable at a, replace $f(a+h)$ by $f(a)+hf\prime(a)$ is in general

A. an exact identity

B. a definition of h

C. a local approximation

D. always false, even for an affine function

## True or false? Justify.

1. A tangent can never meet a curve again.

2. The absolute value function is differentiable at 0.

3. An affine function $mx+p$ has derivative m at every real number.

**Personal review**

I give the domain and h non-zero; I read a slope; I use the point of tangency; I distinguish exact and approximate.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.

# Solutions and teaching notes, chapter 4

**Teacher use.** The following answers concern the tasks not solved in the lesson. Do first make the approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. $\dfrac{8-2}{3-1} = 3$. *Action:* Represent horizontal and vertical increments.

2. $4h + h^2$. *Action:* Expand before simplifying.

3. $4 + h$. *Action:* Factor out h in the numerator before dividing.

4. 0 and 0,1. *Action:* Check by squaring.

5. 3 and 2. *Action:* Represent the distance from 0.

6. $y = 2x + 3$. *Action:* Substitute the coordinates to find the y-intercept.

## Investigation

The slopes are 2 + $h$ ; limit 2. The line passes through (1; 1), so $y = 2x - 1$. The drawing motivates, the justified calculation.

## Direct applications

1. $f(2+h) - f(2) = 7h + h^2$ ; rate 7 + $h$ for $h \neq 0$.

2. The rate is 4 + 3 $h$, so the derivative at this point is 4.

3. $y = 5 - 2(x-3) = -2x + 11$ ; approximation 5 $- 2(0{,}04) = 4{,}92$.

4. The rate at 2 is $\dfrac{|h|}{h}$, with the two opposite one-sided limits. For the square root at 0, the rate $\dfrac{1}{\sqrt{h}}$ for $h > 0$ exceeds every bound as h approaches 0: no real derivative.

## Exercises: methods and answers

**Exercise 4.1 · Two secants.** Slopes 3 and 2.1; derivative 2. The second is closer to the tangent slope; here it is an exact calculation of a rate.

**Exercise 4.2 · A tangent.** Rate 4 $a + 2h = -4 + 2h$, therefore $f'(-1) = -4$. $f(-1)$ = 3 ; tangent $y = 3 - 4(x+1) = -4x - 1$.

**Exercise 4.3 · Marginal cost.** $C(10) = 400$, $C(11)$ = 441, an increase of 41 for one unit. The rate around 10 is 40 + $h$, so the derivative is 40: approximation of the increment for a small change, not an equality for a whole unit.

**Exercise 4.4 · Read a tangent.** On the figure in the lesson: points (1;0) and (2;4), gradient 4. In the new situation, $f(2)$ = 3, slope $-\dfrac{1}{2}$. Approximation 3 $- 0{,}5 \times 0{,}1 = 2{,}95$. The y-coordinate of B is not necessarily $f(4)$.

**Exercise 4.5 · Checking an approximation.** $1+2(-0{,}02)=0{,}96$. Exact 0 ,9604 ; exact minus approximate 0 ,0004 $>0$. The term omitted is $h^2$ : lower approximation.

**Exercise 4.6 · Tangent and intersection.** The tangent has equation $y=1+3(x-1)=3x-2$. Expanding the product does give $x^3-3x+2$. The x-coordinates of the intersection points are therefore 1 and $-2$, by the zero-product rule. The points are (1; 1) and ( $-2;-8$). There are two common points: This is enough to refute the general statement.

**Exercise 4.7 · Numerical secants.** List of rates 2+ $h$ : 2.1 ; 2.01 ; 2.001 and 1.9 ; 1.99 ; 1.999. The algebraic calculation gives the real number 2. For very small h, floating-point calculation may lose precision.

**Exercise 4.8 · Delayed review.** For $h\neq 0,-1$, the rate is $-\dfrac{1}{1+h}$ ; it tends to $-1$. Approximation 1 $-0{,}02=0{,}98$. The exact value $\dfrac{50}{51}$ allows a check.

**CQFD, Exercise 17 p. 127, all questions.** The x-coordinates of A, B and C are 0, 1 and 2. The tangents give slopes 2, 0 and, respectively, $-2$. Thus $f'(0)=2,\ f'(1)=0,\ f'(2)=-2$. The axis units must be read from the grid.

**CQFD, Exercise 19 p. 127, Questions 1 and 2.** Point A is read as (1; 4), so $f(1)=4$. The tangent passes through (0; 1) and (1; 4): slope 3. Thus $f'(1) = 3$ and the tangent has equation $y=4+3(x-1)=3x+1$.

## MCQ: answers and reasons

**MCQ 1.**

1. B. $\dfrac{4-1}{2-1}=3$.

2. A. $4+2(x-1)$.

3. D. It is a value at one point.

4. C. Rate 6 + $h$.

5. B. Affine tangent function.

**MCQ 2.**

1. C. Rate $\dfrac{1}{\sqrt{h}}$ not finite.

2. A. $|h|=-h$.

3. D. Constant y-coordinate.

4. B. A change of position divided by a change of time is expressed in m s $^{-1}$.

5. C. Exact equality for an affine function, approximation in general.

## True or false

1. False: $x^3$ and the tangent 3 $x-2$ at 1 also meet at $-2$.

2. False: the rates on either side are $-1$ and $1$.

3. True: the rate equals $m$ for every nonzero $h$.

**Differentiated teaching.** If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language mastered, then require a precise mathematical sentence in French.